INTERNAL IT OPERATIONS GUIDE

LeaversForm

How the leaver request system works, what each part does, and how to fix it quickly

This guide is written for IT staff who support the application day-to-day. You do not need coding knowledge — think of LeaversForm as a web form plus a scheduled PowerShell offboarding script, connected to Entra ID, Exchange Online, and on-premises Active Directory.

Version: August 2026 (updated) | Organisation: Yours Clothing / SizeUp | Repository: LeaversForm

Contents

1. What LeaversForm does
2. Big-picture architecture
3. End-to-end flow: submit to account disable
4. When does the disable script run?
5. Request statuses explained
6. Using the web application (staff & IT)
7. In-app processing settings & diagnostics
8. Two Entra apps (important)
9. Folder and project structure
10. Test mode vs live mode
11. Development vs production
12. Key configuration files
13. Setup scripts cheat sheet
14. Logs and where to look
15. Quick troubleshooting guide
16. Common problems and fixes
17. Safety rules and go-live checklist

1. What LeaversForm does

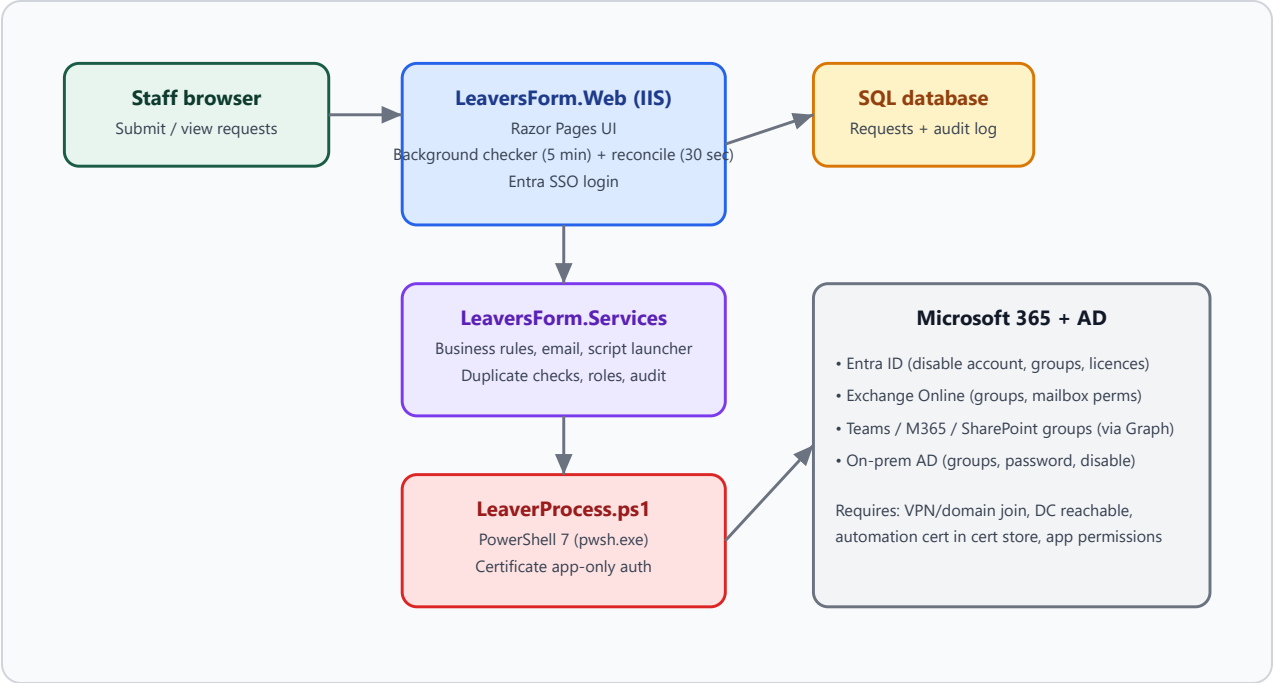
LeaversForm replaces a manual leaver spreadsheet. Managers and HR submit a leaver through a web form. The system:

- Records who is leaving, when, what equipment they have, and which managers approved it
- Emails the selected authorising managers
- At the right time, runs an automated offboarding PowerShell script (`LeaverProcess.ps1`)
- Disables the Entra ID account, disables the on-prem AD account, removes group memberships, revokes licences, and cleans up Exchange/mailbox access where possible
- Shows plain-English status on the request list (Waiting, Processing, Done, Needs IT help)
- Emails IT if something goes wrong

> **Golden rule:** Never run the standalone `LeaversForm.Runner` at the same time as the web app on production. Both would try to process leavers and could cause duplicate runs. The web app already checks every 5 minutes.

2. Big-picture architecture

The system has four main layers. The diagram below shows how they connect.



Web app (LeaversForm.Web)

What users see. Hosted on IIS in production. Includes a background service that polls for due leavers every 5 minutes and reconciles stuck Processing requests every ~30 seconds.

Services layer (LeaversForm.Services)

The "brain": decides when someone is due, launches PowerShell, sends emails, writes audit entries, handles retries.

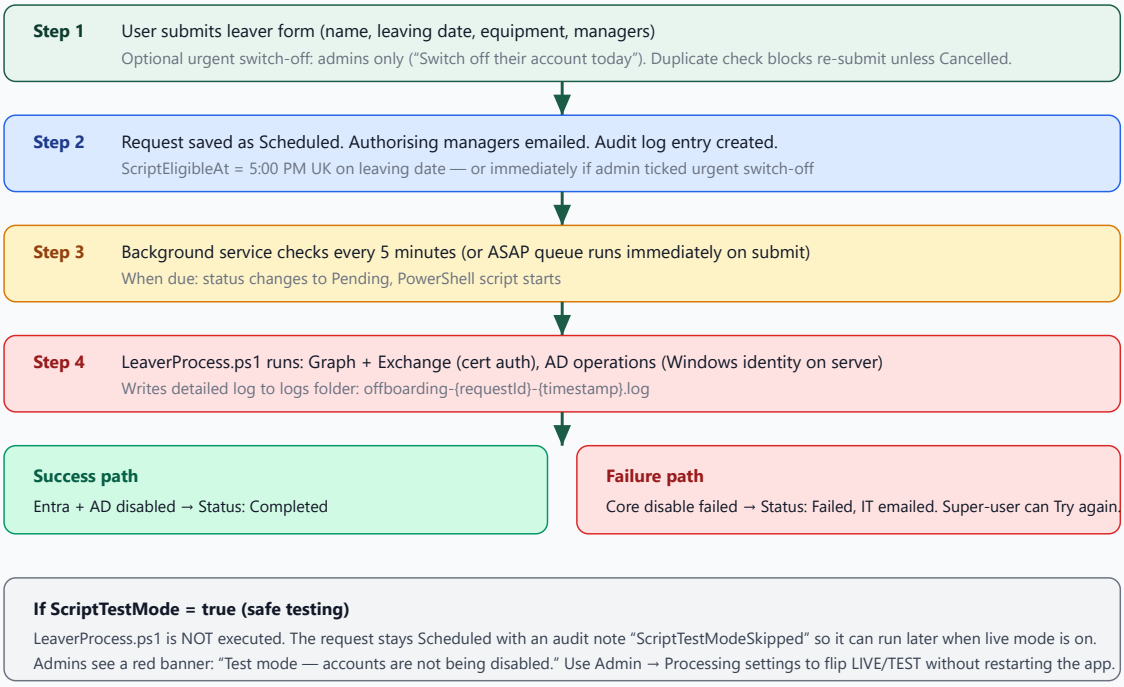
Database (LeaversForm.Data)

Stores requests, statuses, audit trail, authorising managers, and security exclusions. SQLite on dev laptops; SQL Server in production.

Offboarding script (LeaverProcess.ps1)

Does the actual account work. Uses a separate Entra app registration with a certificate — not the same as the website login app.

3. End-to-end flow: submit to account disable



4. When does the disable script run?

Scenario	When script runs	Notes
Normal submit No urgent switch-off	**5:00 PM UK time** on the leaving date (within ~5 minutes after, when the background checker runs)	Uses UK timezone with automatic GMT/BST handling. The web app must be running at that time.
Urgent switch-off (admin/super-user only on submit form)	**Immediately** after form submit (background task, usually seconds)	Checkbox: "Switch off their account today (IT admins only)". Regular users cannot enable this.
Edited leaving date	Recalculated to 5 PM on the new date (unless urgent switch-off was used)	Only editable while status is Waiting/Scheduled (admin rules apply).
Failed request — Try again	Immediately when a super-user clicks Try again	Only Failed requests. Resets to Pending and re-runs script.

> **Important:** This is not a Windows Scheduled Task running once at 5 PM. The web application wakes up every **5 minutes**, looks for any requests whose scheduled time has passed, and processes them. That is why IIS must keep the app pool running (Always On, idle timeout 0).

5. Request statuses explained

Staff see **plain-English labels** in the app. IT still uses the internal names in logs, export, and the database.

What users see	Internal status	Meaning
Waiting	Scheduled	Waiting for 5 PM on leaving date (or about to run)
Processing	Pending	Script is running or was interrupted
Done	Completed	Offboarding finished (Entra + AD disabled at minimum)
Needs IT help	Failed	Script failed — IT notified; super-user can Try again
Cancelled	Cancelled	Request withdrawn — hidden from normal list views

Waiting → Processing → Done

Processing → Needs IT help → Try again → Processing

Internal status	Can edit leaving date?	Can cancel?
Scheduled (Waiting)	Yes (admins)	Yes (admins)
Pending (Processing)	No	Yes (admins — attempts to stop script)
Completed (Done)	No	No
Failed (Needs IT help)	No	Yes (admins)
Cancelled	No	No

6. Using the web application (staff & IT)

The app opens on the **Leaver requests** list (not the submit form). This is intentional — most visits are to check status.

Page / area	Who	What it does
Requests (list)	Everyone with access	See leavers you submitted or authorise. Columns: name, leaving date, status (+ subtitle with timing). Urgent requests show an **Urgent** badge. Admins can **Show more columns** for equipment and submitted-by.
Submit	All signed-in users	Search colleague by name/email, pick leaving date, tick equipment, choose authorising managers (they get an email), submit. Admins/super-users may tick **Switch off their account today (IT admins only)** for immediate processing.
Request details	Submitter, authorising managers, admins	Full info, plain status, equipment notes. Admins see sign-in name (UPN); everyone sees work email.
Admin menu (nav)	Super-user only	Dropdown: **Processing settings**, **Blocked accounts**, **User roles**.
Export Excel	Admin / super-user	On the requests list toolbar — secondary button.

> **Banners users see:**
>
> - **Regular staff:** no test-mode or technical banners (only success messages after actions).
> - **Admins:** one red banner when test mode is on, with a link to Processing settings.
> - **Super-users:** email/sandbox/script-path details live under Processing settings → Technical details (collapsed).

7. In-app processing settings & diagnostics (super-user)

Most day-to-day IT work no longer requires RDP, log folders, or PowerShell CLIs. Super-users use the web UI at **Admin** → **Processing settings** (`/Leavers/Ops`).

Feature	What it does
LIVE / TEST toggle	Emergency stop or go-live without restarting the app. Takes effect immediately and persists `ScriptTestMode` to the local settings file (`appsettings.*.local.json`). Switching to LIVE requires typing `LIVE` . Audit entry: `ProcessingModeChanged` .
Fix stuck requests	Runs reconciliation for all stuck Pending/Scheduled requests (same as automatic 30-second background reconciliation, but on demand).
Fix automation setup	Repairs PowerShell module environment (`Repair-LeaversFormModules.ps1` pattern).
Technical details (collapsed)	Script path, log folder, cert configured, email test mode, AD sandbox skip — for troubleshooting only.

Per-request diagnostics (super-user, on Details page)

Expand **Diagnostics (super user)** on any request:

- **Audit timeline** — every action (Created, ScriptStarted, ScriptCompleted/Failed, Cancel, ProcessingModeChanged, etc.)
- **Platform status** — Graph / Exchange / AD summary from the latest offboarding log
- **Download logs** — offboarding and summary execution logs (path-validated)
- **Fix stuck request** — reconcile one request when status is Processing or stale Waiting
- **Script process active** — yes/no while Processing

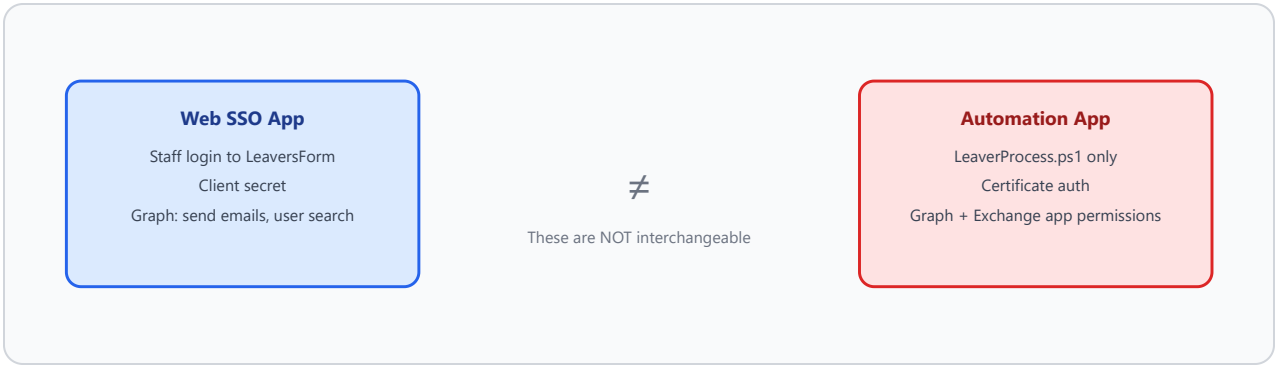
> **Cancel while Processing:** Admins can cancel a Pending request. The app attempts to terminate the in-flight PowerShell process. Always verify account state in Entra/AD after cancelling — termination is best-effort.

> **Automatic reconciliation:** Every ~30 seconds the web app reconciles stuck Pending requests (reads offboarding logs, terminates hung processes, marks Failed or Completed). The in-app “Fix stuck” buttons trigger the same logic immediately.

8. Two Entra apps (important)

LeaversForm uses **two separate Entra ID app registrations**. Mixing them up is the most common certificate/permission problem.

	Web SSO app	PowerShell automation app
Purpose	Users sign into the website	LeaverProcess.ps1 connects to Graph & Exchange
Auth method	Client secret (in app settings)	Certificate (private key in cert store; public .cer uploaded to Entra)
Config keys	`AzureAd:ClientId`, `ClientSecret`	`LeaversForm:PowerShellAppId`, `PowerShellCertificateThumbprint`
Typical mistake	—	Uploading the .cer to the wrong app, or using the .cer file locally (it has no private key — you need the .pfx in the cert store)



9. Folder and project structure

```
LeaversForm/
├── ── Web application (what users open in browser)
├── LeaversForm.Web/          Pages, layout, IIS-hosted background checker
├── LeaversForm.Services/     Business logic, PowerShell launcher, Graph email
├── LeaversForm.Data/         Database models, migrations, seed data
├── LeaversForm.Runner/       Optional fallback batch job (do NOT run with Web in prod)
├── LeaversForm.Tests/        Automated unit tests
│
├── ── Offboarding script
├── LeaverProcess.ps1         Master copy of the disable script
├── scripts/                  Dev/runtime copy (auto-synced by the app)
├── logs/                     Offboarding logs (dev); production uses publish folder\logs
│
├── ── Deployment & operations
├── deploy/                   IIS setup, publish, firewall, cert ACL, readiness tests
├── docs/                     This guide + deployment notes
├── Final-Build.ps1           One-click production server setup (run as admin)
│
├── ── Dev / testing helpers (laptop)
├── Start-LanTest.ps1         Run the site locally over HTTPS on your LAN
├── Set-TestProcessing.ps1    Turn test mode ON (ScriptTestMode true)
├── Set-LiveProcessing.ps1    Turn live script mode ON (dev only — careful!)
├── Initialize-LiveProcessingTest.ps1   First-time laptop setup for real script testing
├── Repair-LiveProcessingCertificate.ps1  Fix cert/thumbprint issues
└── Test-LiveProcessingReadiness.ps1      Preflight checks before going live
```

What each project does (simple view)

Project / folder	Think of it as…
`LeaversForm.Web`	The website + the 5-minute background timer
`LeaversForm.Services`	Rules engine: who can submit, when to run script, send alerts
`LeaversForm.Data`	The database schema and startup seed data (managers list, etc.)
`LeaversForm.Runner`	Standalone “run once and exit” processor — emergency fallback only
`deploy/`	Scripts used when installing on the real server
`LeaverProcess.ps1`	The actual account disable / cleanup automation

10. Test mode vs live mode

Three safety switches control how “real” the system behaves. The most important is `ScriptTestMode`.

Preferred ways to flip test/live (no app restart required):

- **Super-user:** Admin → **Processing settings** → Switch to TEST or LIVE (LIVE requires typing `LIVE`).
- Effect is **immediate** for the running web app. The setting is also saved to `appsettings.Development.local.json` or `appsettings.Production.local.json`.

PowerShell / config file (still supported):

1. `.\Set-TestProcessing.ps1` or `.\Set-LiveProcessing.ps1` / `.\Enable-LiveProcessing.ps1`
2. **Restart** the web app if you only changed the file outside the UI (IIS recycle or stop/start `Start-LanTest.ps1`).

On startup, confirm logs or Processing settings show the expected mode. Admins see a red banner when test mode is on.

How to tell test mode ran instead of the real script:

- Audit log contains `ScriptTestModeSkipped`
- Request stays **Waiting** (Scheduled) — not Done
- No file `logs\offboarding-{requestId}-*.log` with real disable steps

Fix: switch to LIVE in Processing settings (or run live scripts), then submit again with urgent switch-off if testing immediately.

Setting	Test (safe)	Live (production)
`LeaversForm:ScriptTestMode`	**true** — script does NOT run, accounts NOT disabled	**false** — LeaverProcess.ps1 runs for real
`Graph:EmailTestMode`	**true** — emails logged, not sent	**false** — real emails via Microsoft Graph
`Graph:RedirectAllEmailsTo`	Set to a tester mailbox (semi-live / dev) — all mail redirected there	Empty — mail goes to real recipients
`LeaversForm:ScriptSandboxSkipAd`	**true** — AD disable skipped (Entra still runs); dev/sandbox only	**false** — full AD disable

> **Typical dev laptop:** Use Processing settings or `Set-TestProcessing.ps1` when colleagues test the form. Use live mode only when you intend to disable real accounts.

11. Development vs production

Topic	Development (laptop)	Production (server)
How it runs	`Start-LanTest.ps1` or `dotnet run`	IIS website + Always Running app pool
Database	SQLite file `leaversform.db`	SQL Server
Certificate store	CurrentUser\My (laptop testing)	LocalMachine\My + IIS app pool ACL
Script / logs path	`scripts/` and `logs/` in repo folder	Inside IIS publish folder (e.g. `C:\inetpub\LeaversForm\Web\logs`)
Setup script	Initialize-LiveProcessingTest.ps1	Final-Build.ps1 -EnableLiveMode

12. Key configuration files

File	Purpose
`appsettings.json`	Base defaults — test modes ON, safe for fresh install
`appsettings.Development.local.json`	**Gitignored secrets** — Entra IDs, cert thumbprint, ScriptTestMode (dev laptop)
`appsettings.Production.local.json`	**Gitignored secrets** on server — SQL connection, client secret, live flags
`appsettings.LanTest.local.json`	Written by Start-LanTest.ps1 — LAN HTTPS URL and cert path

13. Setup scripts cheat sheet

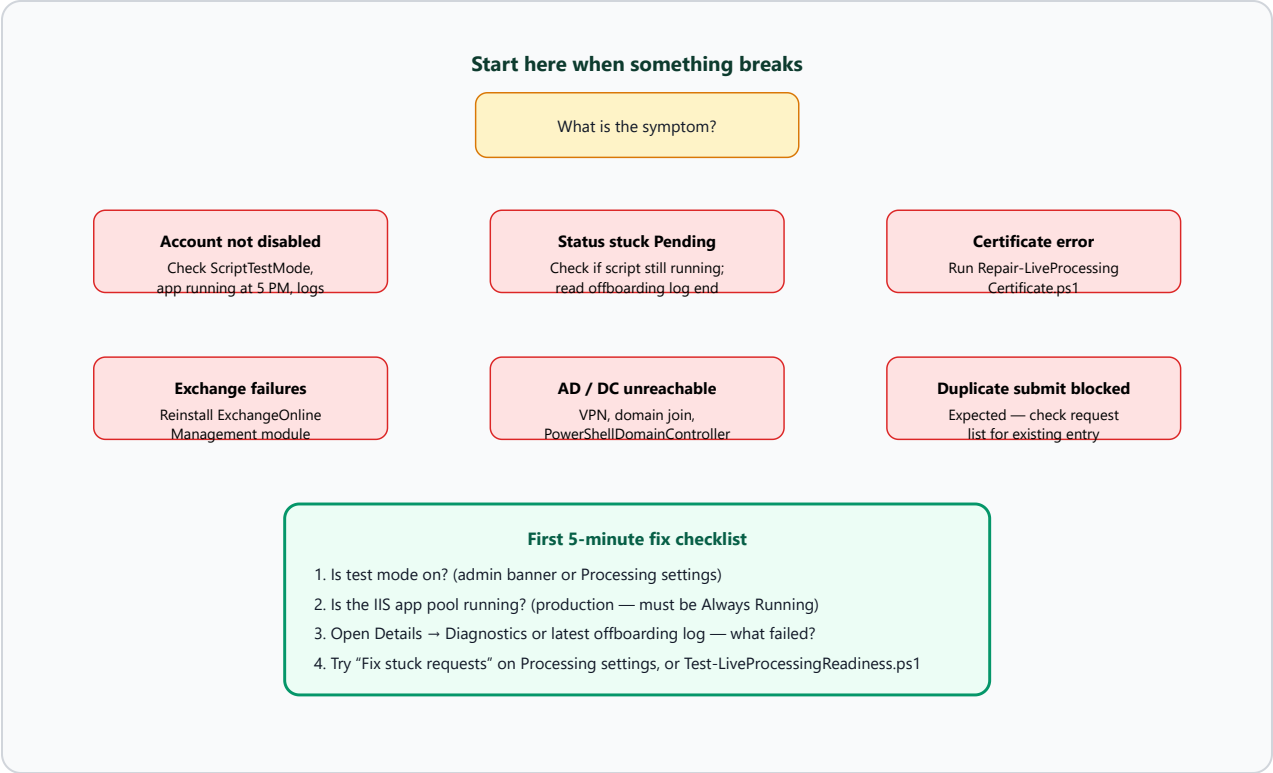
Script	When to use
`Start-LanTest.ps1`	Run the site locally for demos/testing on your LAN
`Initialize-LiveProcessingTest.ps1`	First-time laptop setup: create cert, export .cer for Entra, configure script paths
`Repair-LiveProcessingCertificate.ps1`	"Certificate not found" or wrong thumbprint — imports .pfx and fixes config
`Test-LiveProcessingReadiness.ps1`	Run before going live — checks modules, cert, script path, test mode flag
`Enable-LiveProcessing.ps1`	Quick: ScriptTestMode false + readiness check (then restart app)
`Set-TestProcessing.ps1`	Quick: ScriptTestMode true in local JSON (restart app if not using UI toggle)
`Set-LiveProcessing.ps1`	Full live setup wizard (cert, paths, ScriptTestMode false)
`scripts\reset-leaver-requests.ps1`	Remove test/false requests: `-List`, `-Name Bachu`, `-Id 18`, `-All`
`Final-Build.ps1 -EnableLiveMode`	Production server: publish, cert, IIS, modules, checklist
`Repair-LeaversFormModules.ps1`	Same as in-app "Fix automation setup" — verify/repair Graph + Exchange modules
`LeaversForm.Runner --reconcile`	CLI fallback to fix stuck requests (prefer Admin → Processing settings)
`deploy\Test-ProductionReadiness.ps1`	Post-install checks on the server

14. Logs and where to look

Log	Location	Contains
Offboarding log (most useful)	`logs\offboarding-{requestId}-*.log`	Step-by-step script output: Graph connect, AD disable, group removals, errors
App execution summary	`logs\{requestId}_{timestamp}.log`	Exit code + link to offboarding log excerpt
IIS / application	Windows Event Viewer → Application; IIS stdout logs if enabled	Web app crashes, startup errors, ScriptTestMode banner
Audit trail (in app)	Request Details → Diagnostics (super-user) — or cancel/export audit in DB	Created, ScriptStarted, ScriptCompleted/Failed, ScriptTestModeSkipped, ProcessingModeChanged, Cancel, Try again
In-app log download	Details → Diagnostics → Download offboarding / summary log	Same files as on disk; super-user only

> **Tip:** When IT gets a failure email, open the latest `offboarding-{id}-*.log` first. Search for `[ERROR]` or `Failed:`. The Entra + AD disable lines are the most important for confirming core success.

15. Quick troubleshooting guide



16. Common problems and fixes

Problem: Graph "assembly already loaded" / Get-MgUser fails

- **Automatic:** Each run of `LeaverProcess.ps1` now aligns Graph module versions, repairs if needed, and restarts in a fresh PowerShell session if required. You should not need manual `Uninstall-Module` steps.
- **Manual (optional):** Run `.\Repair-LeaversFormModules.ps1` once if you want to verify modules outside a leaver run.
- Ensure `LeaversForm-ModuleBootstrap.ps1` sits beside `LeaverProcess.ps1` in the scripts folder (synced automatically on app startup).

Problem: Status Completed but account not disabled (empty offboarding log)

- **Cause (current builds):** Unlikely if status shows Done — check audit for `ScriptTestModeSkipped` (request would still be Waiting).
- If status is Done but account live: read offboarding log — Entra/AD lines must show disabled.
- Confirm Processing settings shows LIVE, or run `.\Enable-LiveProcessing.ps1`.
- Super-user: Try again on Failed, or cancel and re-submit with urgent switch-off for a controlled retest.

Problem: Someone submitted a leaver but nothing happened at 5 PM

- Check `ScriptTestMode` — if true, nothing will disable (by design).
- Confirm the web app / IIS site was running at 5 PM. If the app pool was stopped, processing resumes when it restarts.
- Open the request on the list — is status still **Scheduled**? Check leaving date is correct.
- Look in `logs\` for any new files after 5 PM.

Problem: Request stuck on Processing

- Check latest `offboarding-{id}-*.log` — does it end with "Finished offboarding"?
- Wait up to ~30 seconds — automatic reconciliation runs in the background.
- Super-user: Details → **Fix stuck request**, or Processing settings → **Fix stuck requests**.
- If genuinely stuck > 45 minutes with no log progress, status may move to Needs IT help — Try again.
- CLI fallback: `LeaversForm.Runner --reconcile`

Problem: Certificate / "private key not found"

- The `.cer` file is for Entra upload only — it cannot run the script locally.
- Run `.\Repair-LiveProcessingCertificate.ps1` to import the `.pfx` from `.live-processing-test\`.
- On production: confirm cert is in LocalMachine\My and IIS app pool has read access (`deploy\Grant-LeaversFormCertificateAccess.ps1`).
- Verify thumbprint in config matches: `Get-ChildItem Cert:\LocalMachine\My` (or CurrentUser\My on laptop).

Problem: Graph connects but Exchange fails

- Common on dev laptops: "Module could not be correctly formed" — reinstall ExchangeOnlineManagement module.
- Entra + AD disable can still succeed; Exchange cleanup (DLs, shared mailboxes) needs EXO working.
- Confirm automation app has Exchange Online application permissions + admin consent.

Problem: AD user not found / DC unreachable

- Server or laptop must reach the domain controller — VPN if off-site.
- Check `LeaversForm:PowerShellDomainController` (e.g. `SH-DC-01.sizeup.local`).
- On production, IIS app pool should run as a domain service account with AD delegate permissions — not ApplicationPoolIdentity.

Problem: "User already submitted" when creating request

- Expected behaviour — prevents duplicate leaver entries for the same person.
- Check the request list for an existing Scheduled, Pending, Failed, or Completed entry.

- Only **Cancelled** requests allow the same person to be submitted again.

Problem: IT failure email received

- Super-user: open request Details → expand Diagnostics (audit + platform status + download logs).
- Read `offboarding-{id}-*.log` for the real error (email includes a platform summary).
- Fix root cause, then super-user clicks **Try again** on Failed requests.

17. Safety rules and go-live checklist

> **Before letting anyone use the form in production:**
>
> - `ScriptTestMode` must be **false**
> - Automation certificate uploaded to the **PowerShell app** (not web app) in Entra
> - Graph + Exchange application permissions granted with admin consent
> - IIS app pool: Always Running, idle timeout 0, preload enabled
> - Domain service account with cert private key access + AD permissions
> - Run one controlled urgent switch-off test against a test account first (admin submits with "Switch off their account today")
> - Confirm Processing settings shows LIVE and no admin test banner
> - Super-users configured on User roles page (or verify seed accounts)

Roles in the application

Role	Can do
User	Submit leavers; view own requests and requests where they are an authorising manager (read-only)
Admin	View all requests; edit Waiting requests; cancel Waiting/Processing/Failed; export Excel; tick urgent switch-off on submit
Super-user	Everything admin can do + Try again on Failed + Processing settings (LIVE/TEST, fix stuck, repair modules) + Blocked accounts + User roles + per-request Diagnostics

Security exclusions ("Blocked accounts")

Super-users maintain rules that prevent certain accounts from being submitted (e.g. ADM_ prefix, named executives, CST in name). Matches are enforced at user search and submit time.

What LeaverProcess.ps1 removes (summary)

- Entra: group memberships (Teams, M365, SharePoint-backed, security, mail-enabled), admin roles, app role assignments, licences, sign-in sessions, account disabled
- Exchange (when connected): distribution groups, mailbox forwarding, Full Access / Send As / Send on Behalf, auto-replies
- On-prem AD: group memberships, dial-in setting, password reset, account disabled

> **Rollback to safe mode:** Admin → Processing settings → Switch to TEST, or run `.\Set-TestProcessing.ps1` . No accounts will be disabled until switched back to LIVE.

LeaversForm IT Operations Guide — August 2026

See also: `docs/PRODUCTION-DEPLOYMENT.md` , `docs/SEMI-LIVE-TESTING.md` , `README.md`

For questions about this system, contact the LeaversForm project owner / internal IT development team.